3.8 MDSD device

3.8.1 General

CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!**

The MDSD device is responsible for driving the horizontal movement of the ptab. Two types of MDSD are introduced.

MDSD for whole body movement.

With revision level 2 and higher, the grounding shield and the external filter must not be used anymore. On older systems, the grounding shield and/or the external filter may be installed. Partial drive is not affected.

MDSD for partial drive, not valid for Trio A Tim Systems.

Each of the MDSD device can be replaced**, they are n o t compatible**.

3.8.2 Tools and time required

3.8.2.1 Time

Approx. 120 min

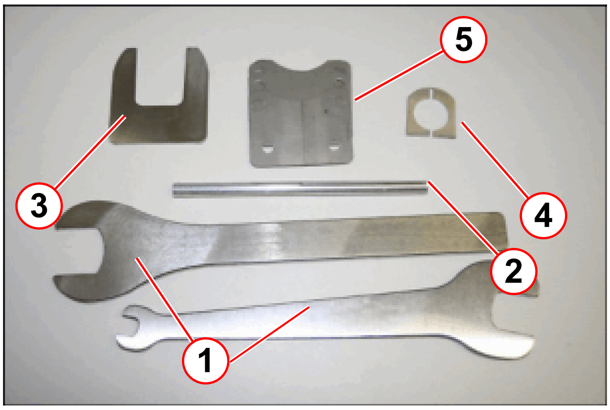
3.8.2.2 Tools

Non-magnetic tool kit

Philips screwdriver

Service aid MDSD (delivered with the MDSD spare part)

Fig. 139: MDSD service aid



(1) Spanners
(2) Feed-in tool for fiber optic cable
(3) Clamp claw
(4) Clamp collar
(5) Shim plates

> For patient tables with partial drive MDSD A4330 (p.n. 10277315), the only service tool included in the spare part is the feed-in tool fiber optical cable. Only for the whole body MDSD A4320 (p.n. 8112240) a complete service aid set, as shown in the figure above, is needed and included in the spare part.

3.8.3 Preliminary

- Position patient table at home position
- Switch OFF F22 (RF-room) in the ACC

> According to the system/room specifications the movement range of the horizontal movement can be reduced.

With the whole body version of the horizontal drive, the movement range of the horizontal movement may be reduced. If that is the case, remove the end stops.
(➔ Fig. 289 Page 173)

Before disassembly the patient table covers on a system with a partial drive unit: The toothed belt must be released to push the table top to the service end of the magnet. Therefore refer to (➔ Prerequisite for cover removal / partial drive / Page 98), not valid for Trio A Tim Systems

3.8.4 Disassembly

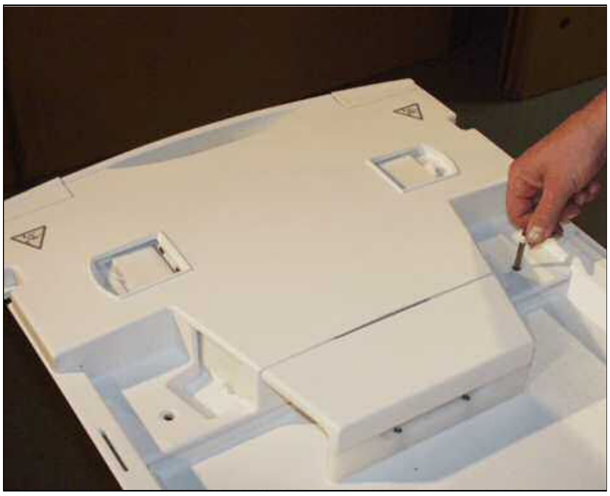
3.8.4.1 Prerequisite for cover removal / partial drive

> Be careful and avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

The tabletop plate may fall - support the tabletop plate.

Basic table

Fig. 140: Coil plug cover (foot end)



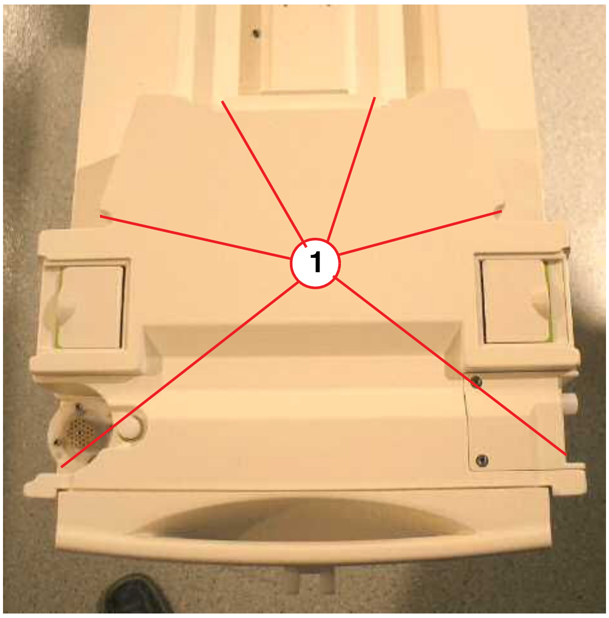
- Remove the Allen screws on top of the cover from the coil plugs.

Fig. 141: Remove Allen screws

- Remove the lower Allen screws of the cover from the coil plugs.

Removable table

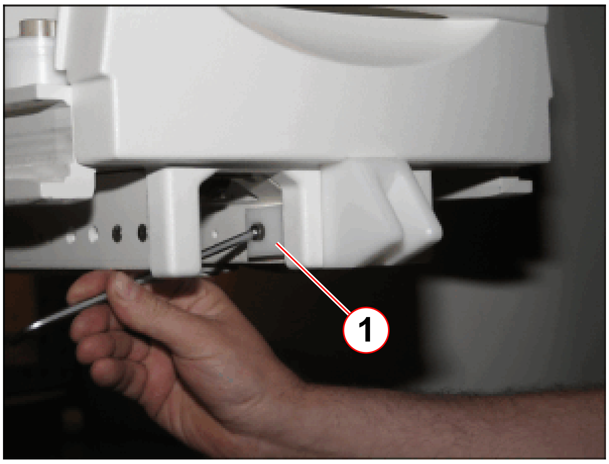
Fig. 142: RF-Cover RTT



(1) Screws to be removed

- Remove the carrier plate of the RTT.
- Remove six screws on top of the cover from the coil plugs.
- Disconnect cables.

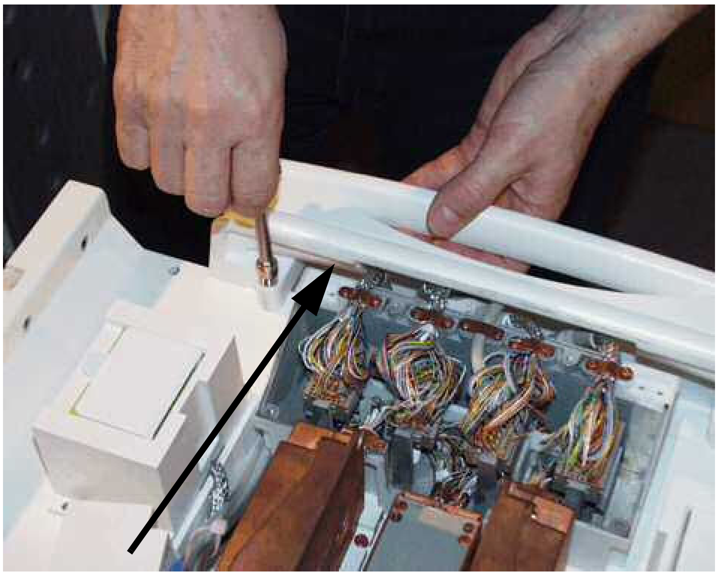
Fig. 143: Additional mounting bracket for foot end cover (newer systems)



(1) Allen screw

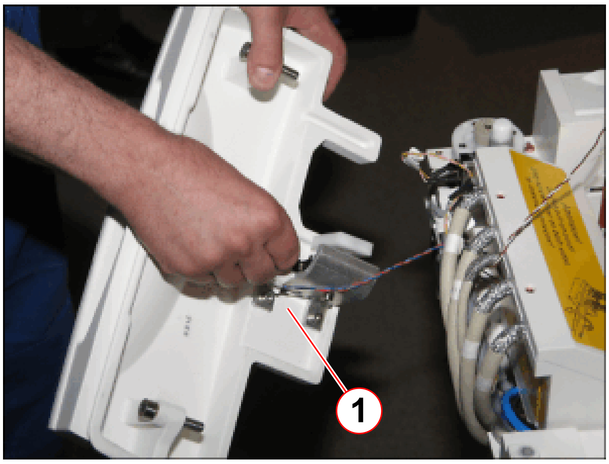
- On newer systems, remove Allen screw of mounting bracket.

Fig. 144: Foot-end cover



- Remove screws of the foot-end cover.

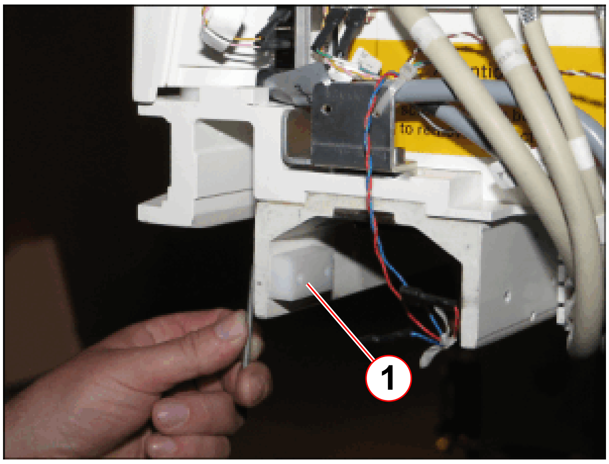
Fig. 145: Foot end cover, removed



(1) Disconnect cables

- Disconnect cable connections and remove the foot-end cover..
- Slide out the lower cover on the bottom of the patient table.

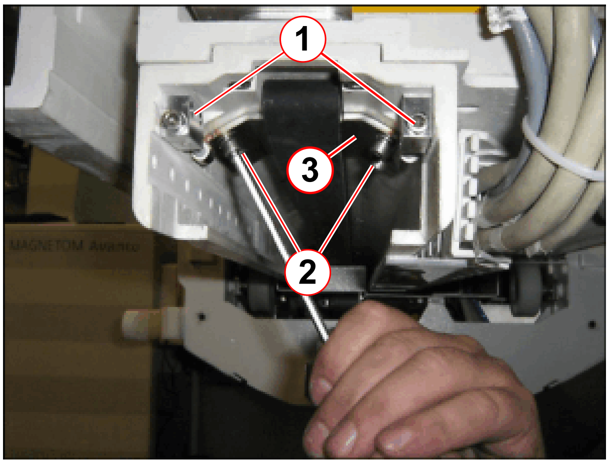
Fig. 146: Mechanical end stop of partial drive unit



(1) End stop to be removed

- Remove mechanical end stop.

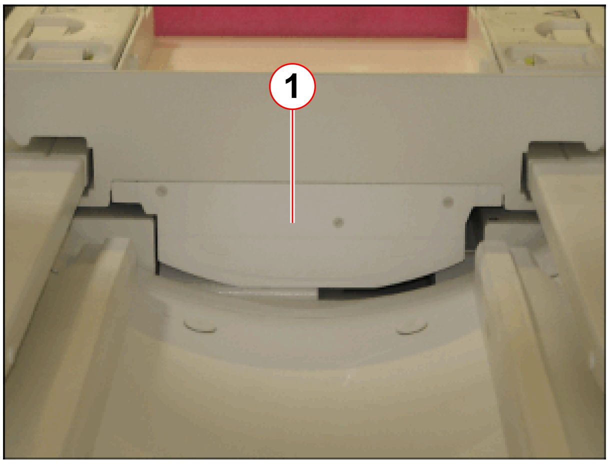
Fig. 147: Tension unit for toothed belt



(1) Allen screws for adjusting belt tension
(2) Attachment screws
(3) clamping plate

- Release tension of toothed belt (1), remove Allen screws (2) and tooth belt tension unit, not valid for Trio A Tim Systems.

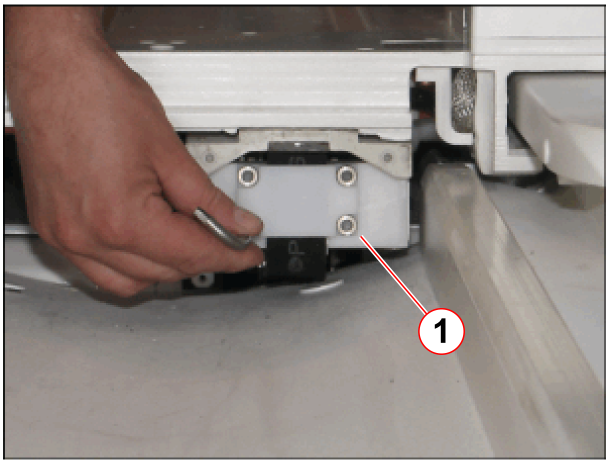
Fig. 148: Cover of toothed belt mount (head end)



(1) Cover of toothed belt mount

- Remove the cover of the toothed belt mount of toothed belt (1) on the head end from the ptab, not valid for Trio A Tim Systems.

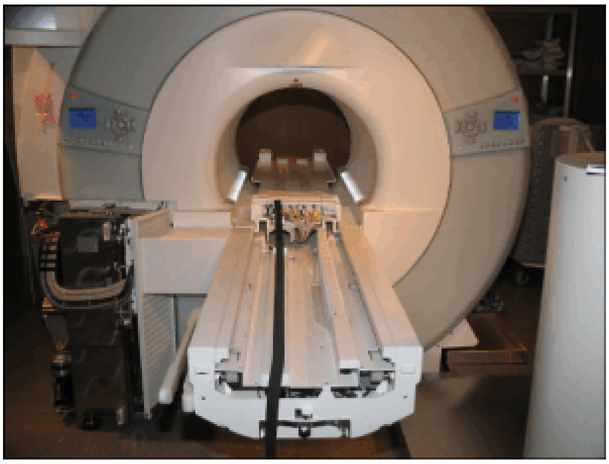
Fig. 149: Toothed belt attachment (head end)



(1) Allen screws of toothed belt mount

- Remove the toothed belt mount (1) on the head end from the ptab, not valid for Trio A Tim Systems.

Fig. 150: Tabletop moved into the magnet



- Carefully push the tabletop plate manually into the magnet.

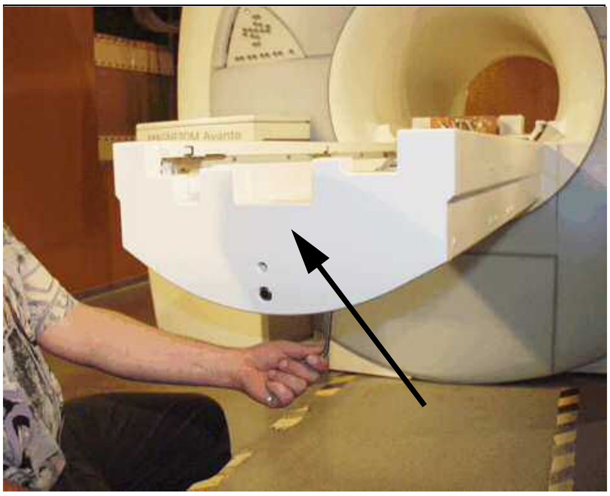
3.8.4.2 Covers

> Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

> For disassembling/assembling the patient table covers we recommend a mid-sized Philips screwdriver.

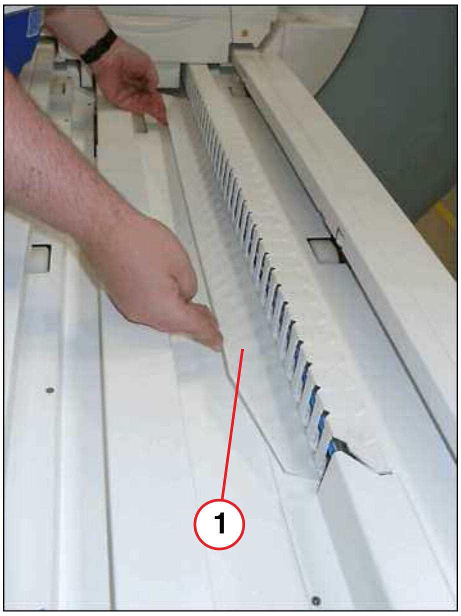
- Push the tabletop plate manually into the magnet (whole body movement).

Fig. 151: Foot-end cover



- Remove the foot-end cover of the patient table.

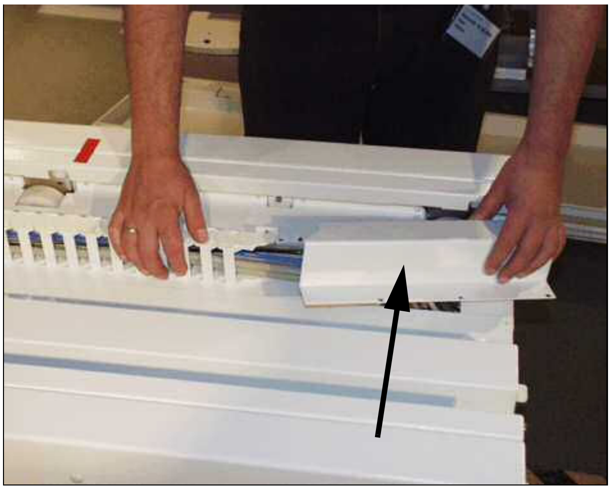
Fig. 152: Plastic inner cover



(1) Plastic inner cover

- Remove inner plastic cover.

Fig. 153: Remove cover of energy chain



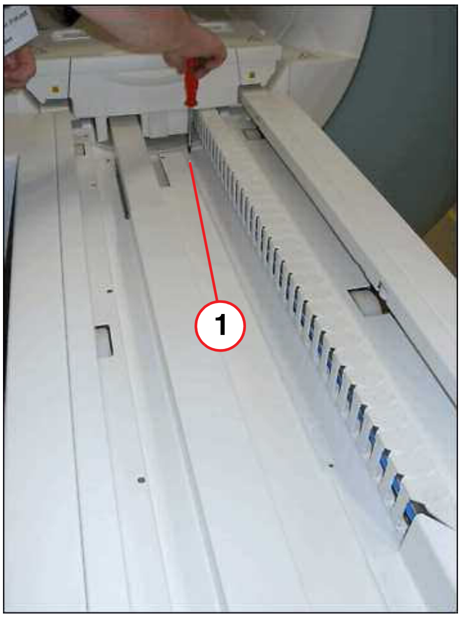
- Remove the upper energy chain cover.

Fig. 154: Remove energy chain

- Remove the upper energy chain of the bracket.

- Remove right guide rail cover (do not remove screws, only loosen).

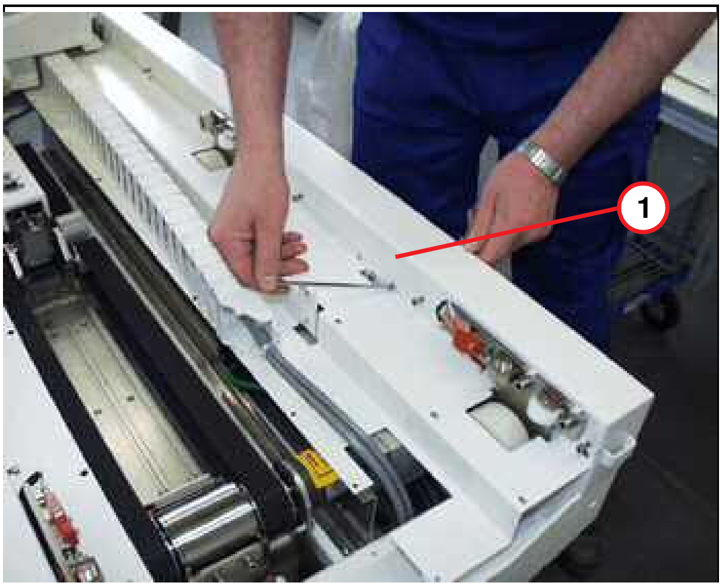
Fig. 155: Inner main cover



(1) Screws to be removed

- Remove screws of main inner cover.

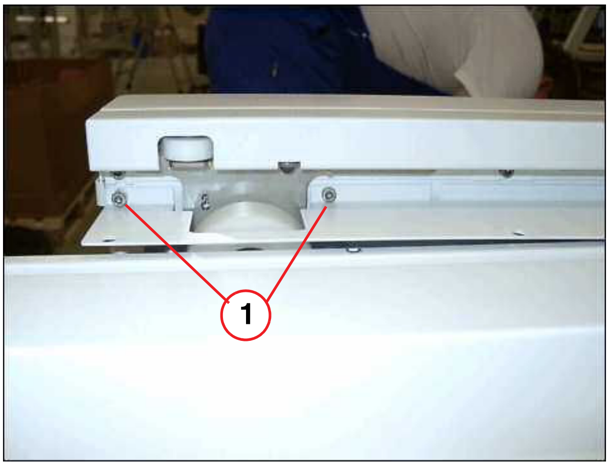
Fig. 156: Inner cover



(1) Top inner cover

- Remove right inner cover.

Fig. 157: Allen screws

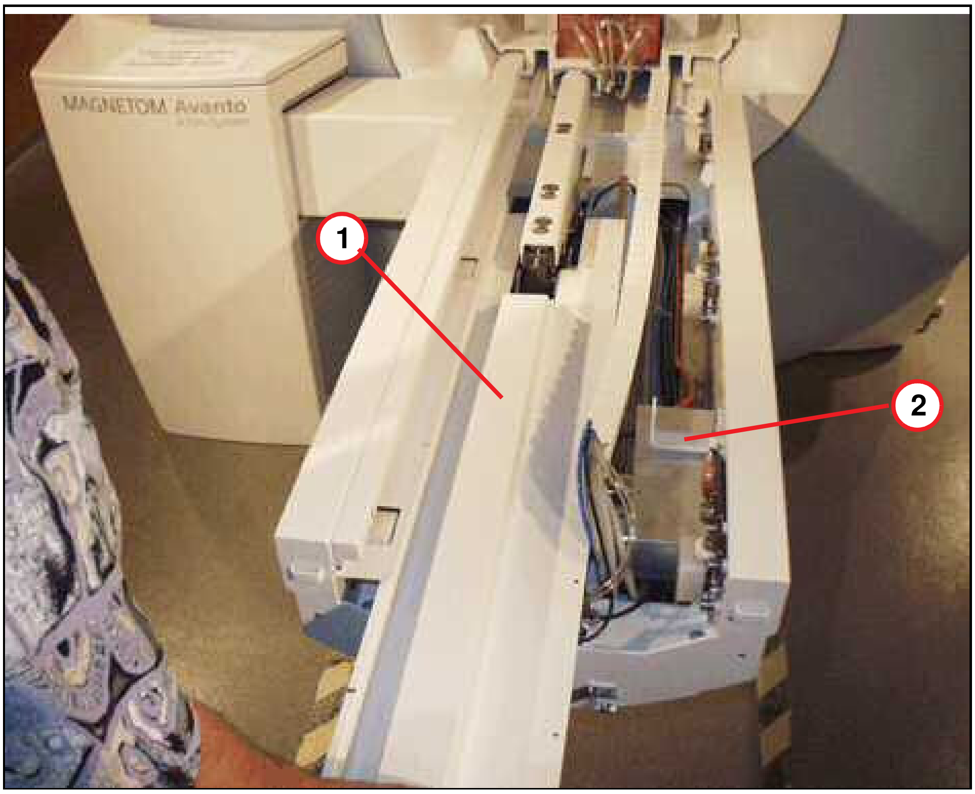


(1) Screws to be removed

- Remove two Allen screws on the left side on the table end (foot) before you remove the main inner cover.

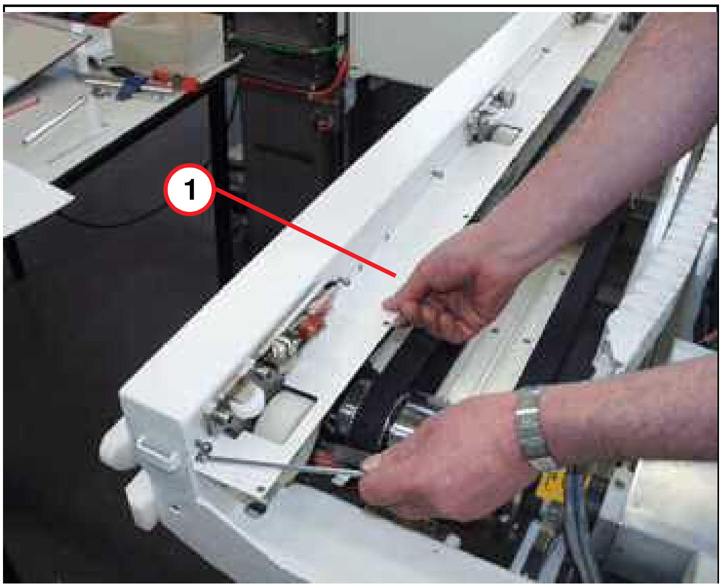
- Remove bracket above the MDSD device.
- Remove main inner cover.

Fig. 158: Disassembling



(1) Main inner cover
(2) Mounting bracket for energy chain

Fig. 159: Inner cover, left side



(1) Inner cover, left side

- Covers removed.

3.8.4.3 MDSD device of whole body drive

Fig. 160: Example - removing FOCs from MDSD

(1) Cam of FOC connector
(2) Screwdriver, flat blade

- The figure below shows the FOC plug connection (principle) inside the MDSD device. Use a flat blade screwdriver to disconnect the FOCs inside the MDSD device.

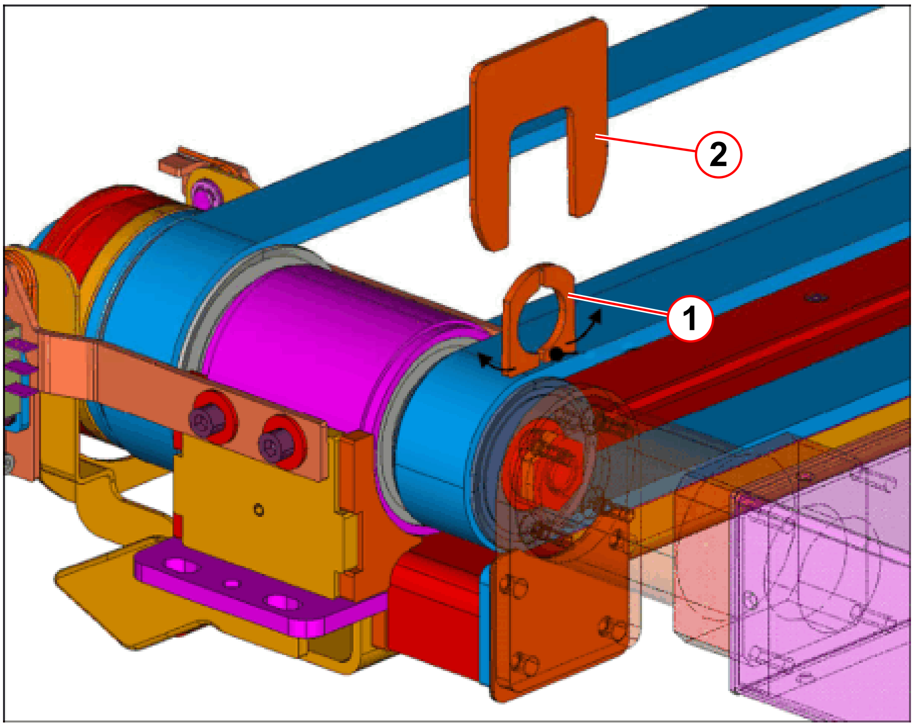
Fig. 161: Fiber-optic cables of MDSD

- Feed the flat blade screwdriver through the hole in the aluminum cover and into the FOC's plug connection. Try to reach the cam of the FOC and carefully push it sideways while at the same time pulling it out from the FOC. Refer to illustration.

- Release the horizontal emergency movement, enabling manual horizontal movement.

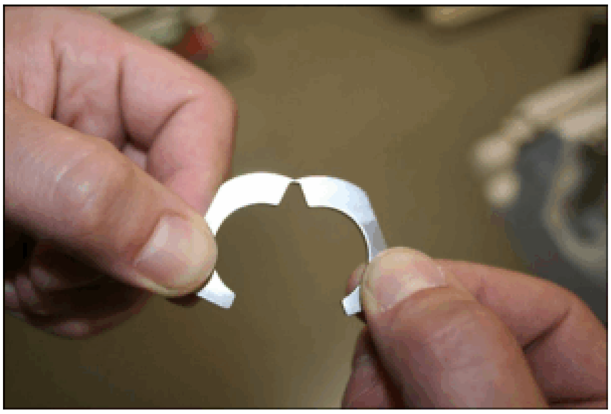
> To avoid the overwinding of the MDSD clamping nut, the clamp collar and the claw fastener (part of the service aid) must be used as described.

Fig. 162: MDSD service aid, principle of use



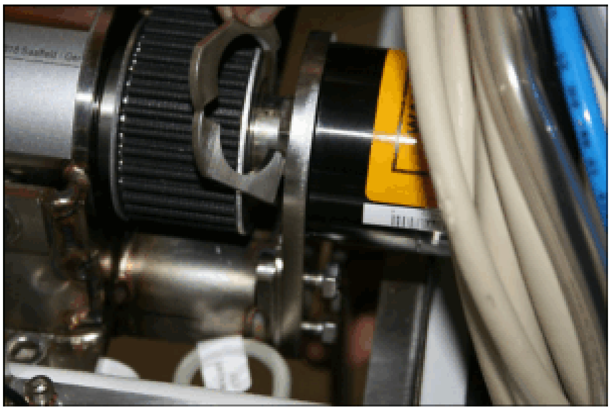
(1) Clamp collar
(2) Claw fastener

Fig. 163: Clamp collar



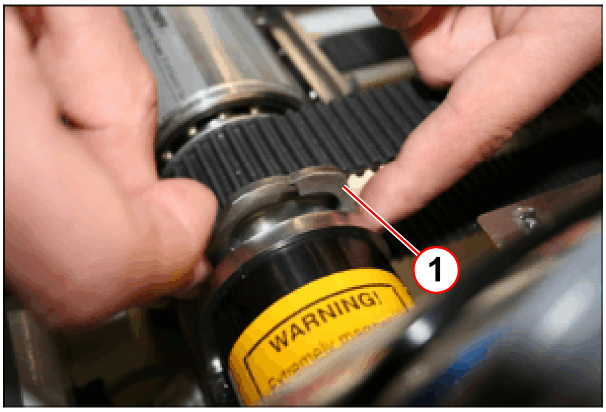
- Carefully bend up the clamp collar, as shown.

Fig. 164: Clamp collar



- Insert the clamp collar.

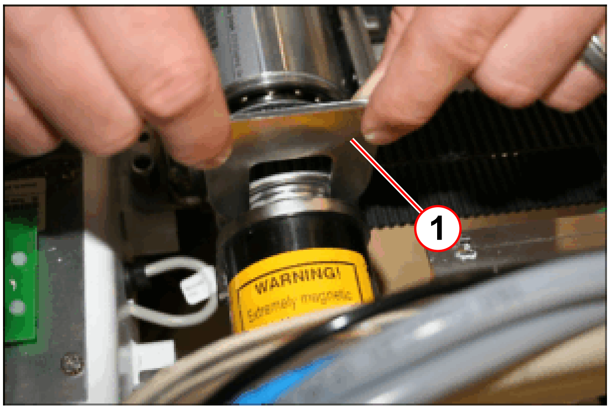
Fig. 165: Clamp collar



(1) Insert clamp collar

- Press the clamp collar together.

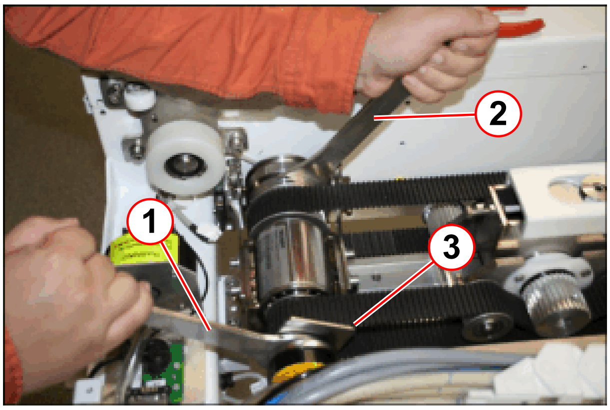
Fig. 166: Claw fastener



(1) Insert claw fastener over the clamp collar.

- Insert the claw fastener (1) over the clamp collar.

Fig. 167: RELEASE or TIGHTENING of the clamping nut



(1) RELEASE, pulling the spanner toward the foot end, TIGHTEN, pushing toward the magnet
(2) Holding the spanner tight
(3) Claw fastener

- Use the open-end spanners of the MDSD service aid to release the MDSD clamping nut.
- **RELEASE,** pulling the open-end spanner (1) **toward the foot end,** while holding the open-end spanner (2).
- The claw fastener should remain.

Fig. 168: RELEASING or TIGHTENING the clamping nut

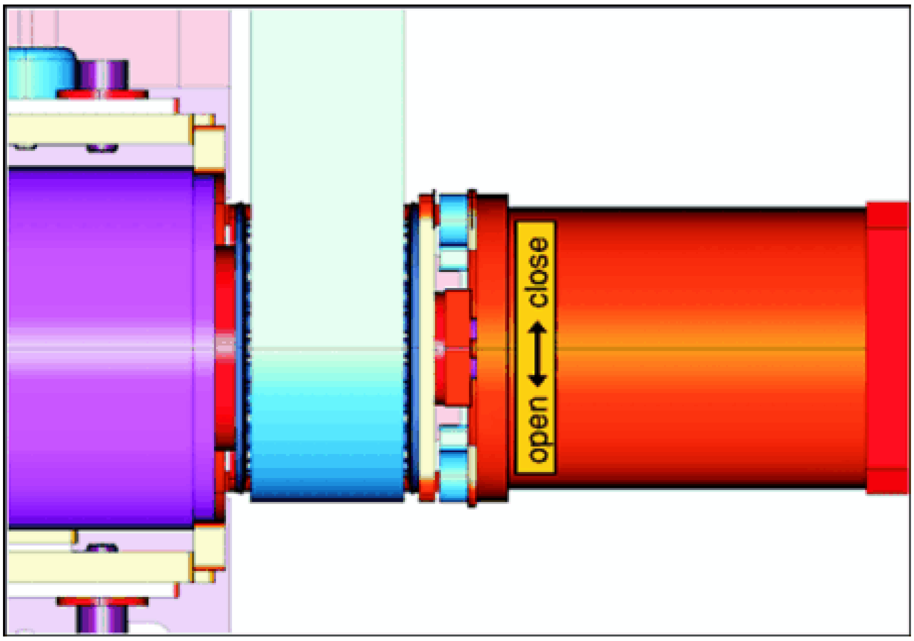
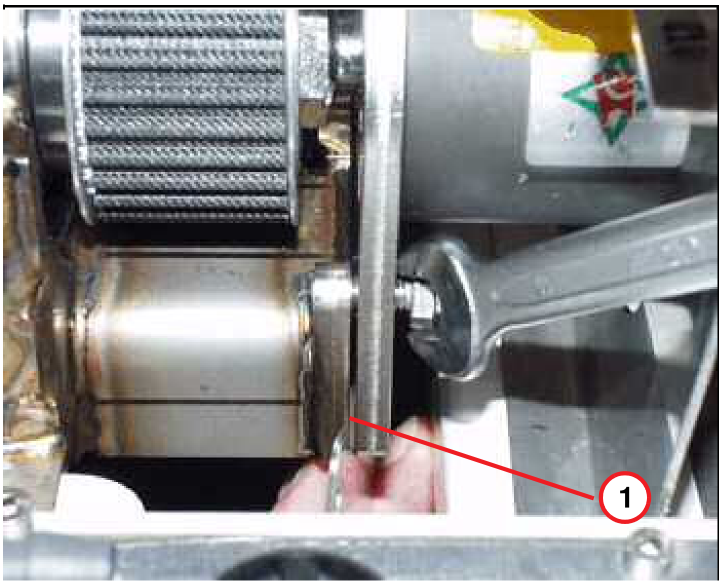


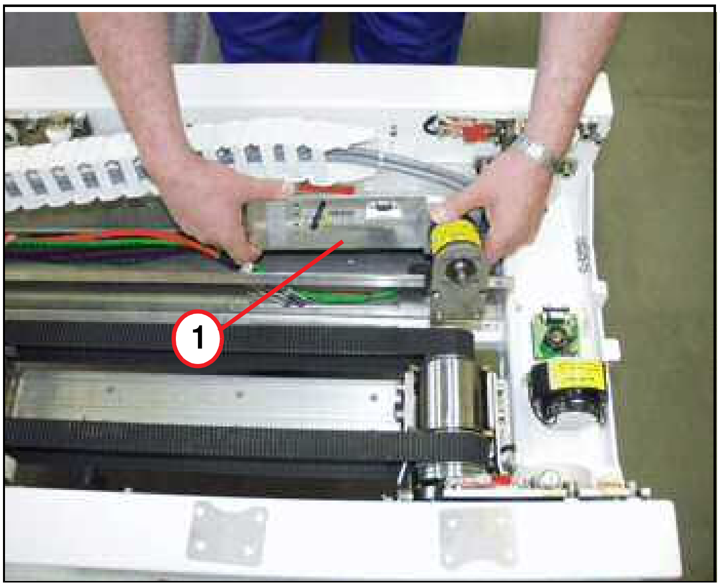
Fig. 169: M6 hex screws of MDSD bracket



(1) Shim gap appears when securing the 22 mm nut

- Remove the four M6 hex nuts.

Fig. 170: *MDSD device*



(1) Replace MDSD device

- Replace the MDSD drive, take care of the correct version.
- Remove the MDSD service aid (clamp collar and claw fastener).

CAUTION

The MDSD drive consists of magnetic objects!

Magnetic objects may cause injury.

» **Maintain a safe distance to the magnetic field!**

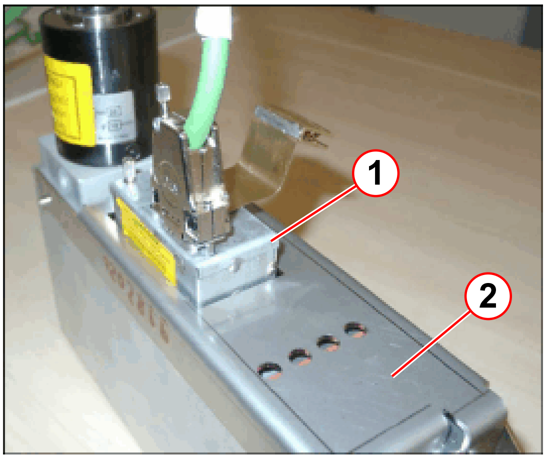
CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!**

Fig. 171: MDSD with external filter and grounding shield



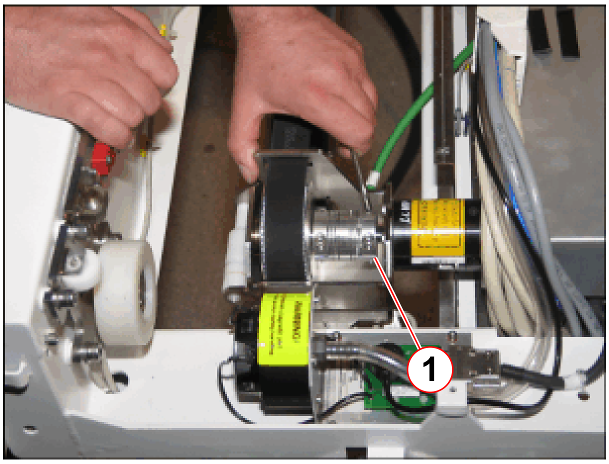
(1) External filter
(2) Grounding shield

Note:

If a grounding shield and/or an external filter is installed, **do not use them anymore for the new MDSD** (applies to revision level 02 and higher).

3.8.4.4 MDSD device of partial drive, not valid for Trio A Tim Systems

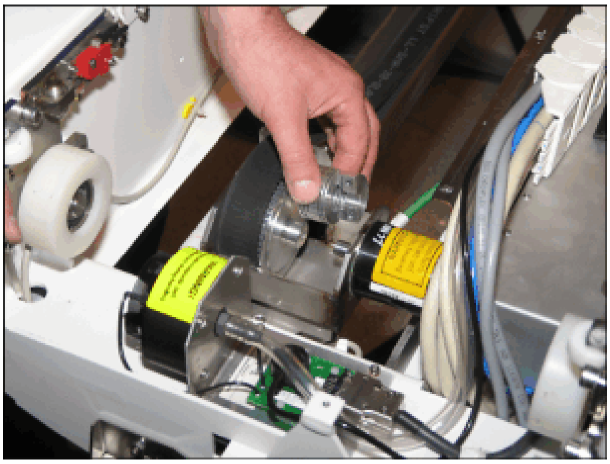
Fig. 172: Clutch of MDSD, partial drive



(1) Allen screws of MDSD clutch

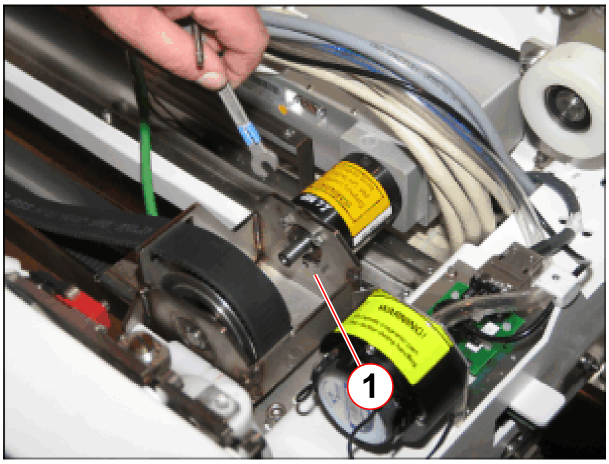
- Remove Allen screws of MDSD clutch (1).

Fig. 173: Clutch of MDSD removed



- Remove MDSD clutch.

Fig. 174: MDSD clutch removed



(1) Mounting screws of MDSD

- Remove the hex screws (1) of the mounting bracket from the MDSD.

Fig. 175: Removing the MDSD

- Replace the MDSD drive.
- Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!

CAUTION

The MDSD drive consists of magnetic objects!

Magnetic objects may cause injury.

» **Maintain a safe distance to the magnetic field.**

CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!**

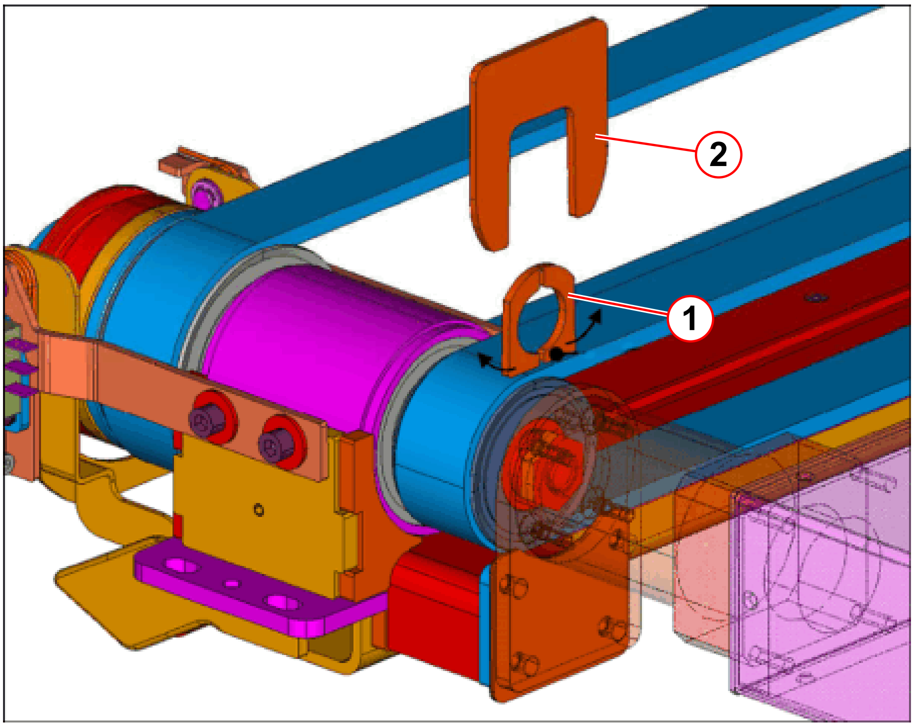
3.8.5 Assembly

3.8.5.1 MDSD drive of whole body drive

The service aid for the MDSD is a prerequisite for assemly.

To avoid the overwinding of the MDSD clamping nut, the clamp collar and the claw fastener (part of the service aid) must be used as described.

Fig. 176: MDSD service aid, principle of use

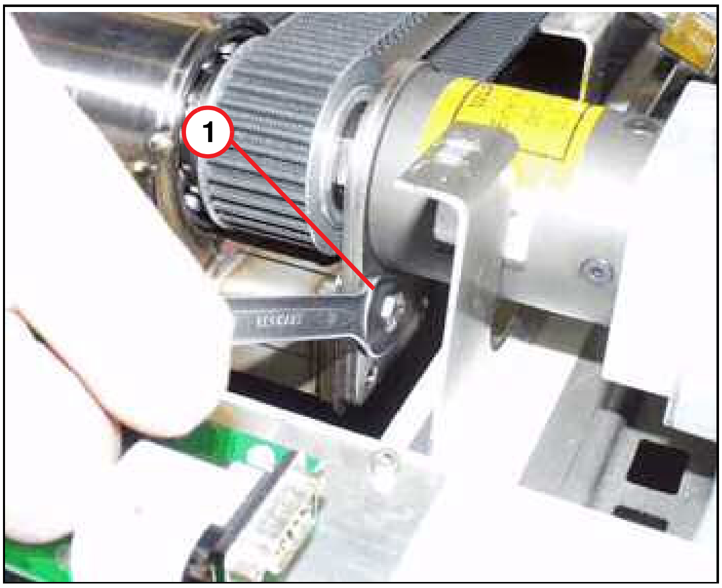


(1) Clamp collar
(2) Claw fastener

Fig. 177: Set of shim plates MDSD drive

- Shim plates must be used when assembling the MDSD device.

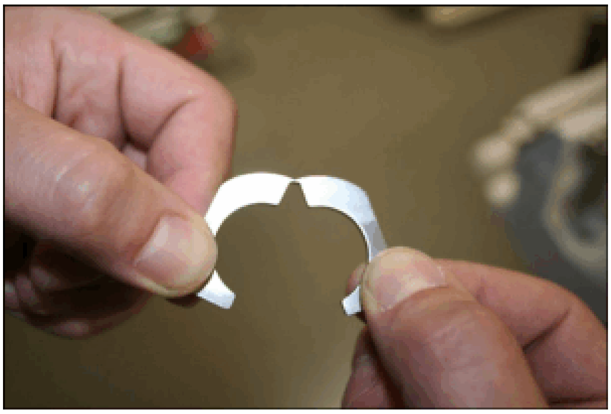
Fig. 178: Hex screws of MDSD



(1) M6 hex screws

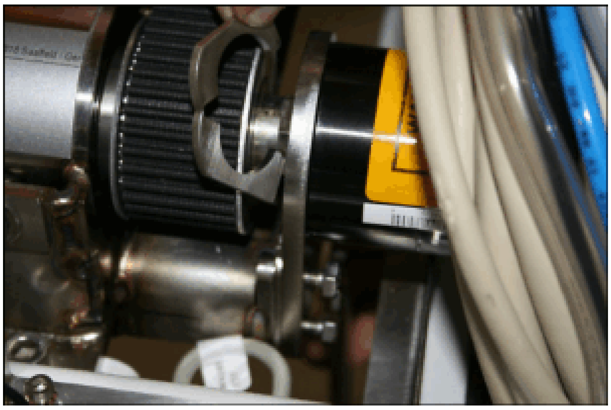
- Insert only the upper two M6 hex nuts and don't tighten them, use them only to hold (as a torque bracket).

Fig. 179: Clamp collar



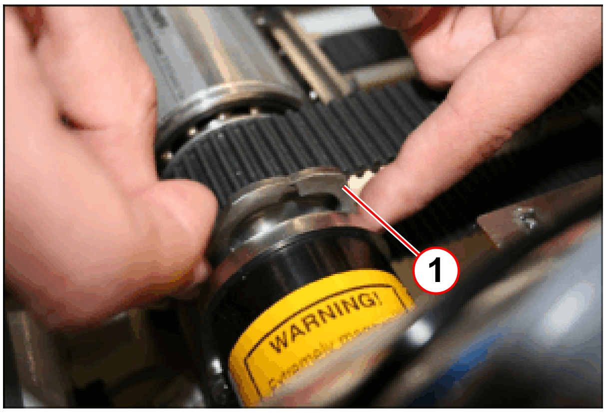
- Carefully bend up the clamp collar, as shown.

Fig. 180: Clamp collar



- Insert the clamp collar.

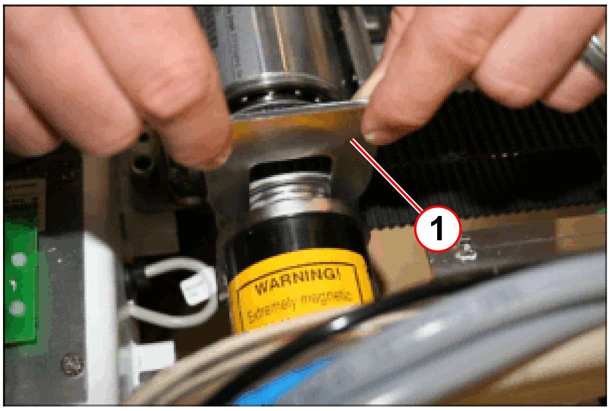
Fig. 181: Clamp collar



(1) Insert clamp collar

- Press the clamp collar together.

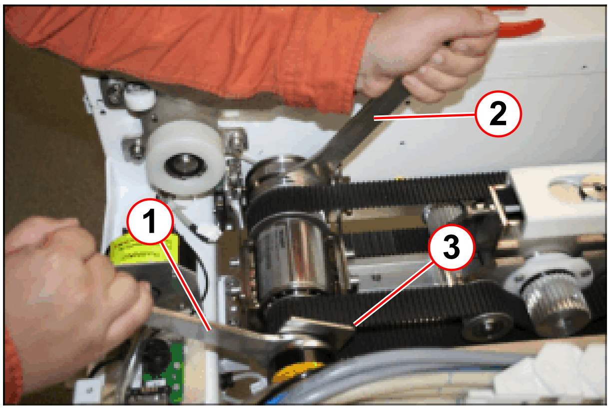
Fig. 182: Claw fastener



(1) Insert claw fastener over the clamp collar.

- Insert the claw fastener (1) over the clamp collar.

Fig. 183: RELEASE or TIGHTENING of the clamping nut



(1) RELEASE, pulling the spanner toward the foot end, TIGHTEN, pushing toward the magnet
(2) Holding the spanner tight
(3) Claw fastener

- Use the open-end spanners of the MDSD service aid to tighten the MDSD clamping nut.
- **TIGHTEN,** pushing the open-end spanner (1) **to the magnet,** while holding the open-end spanner (2).
- The claw fastener should remain.

Fig. 184: RELEASING or TIGHTENING the clamping nut

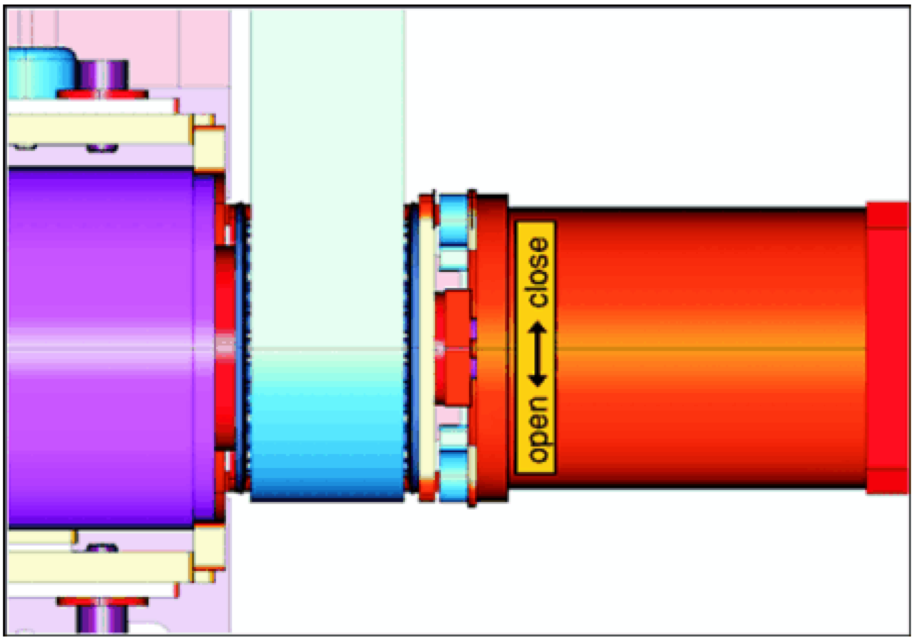
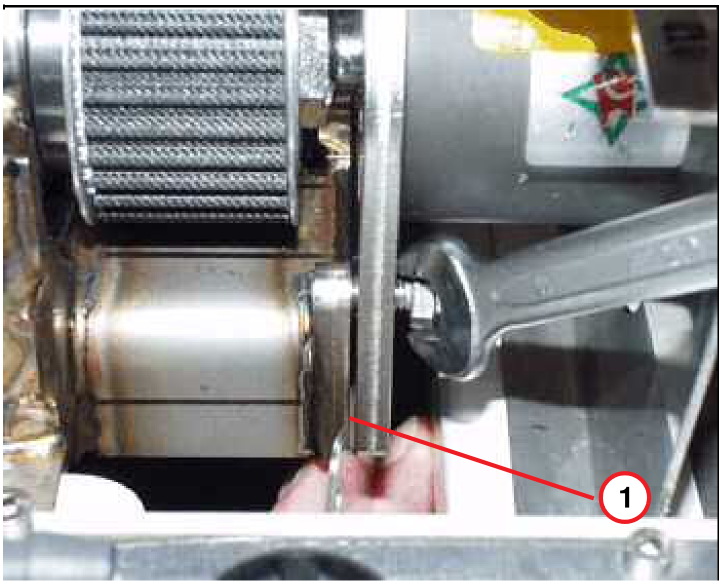


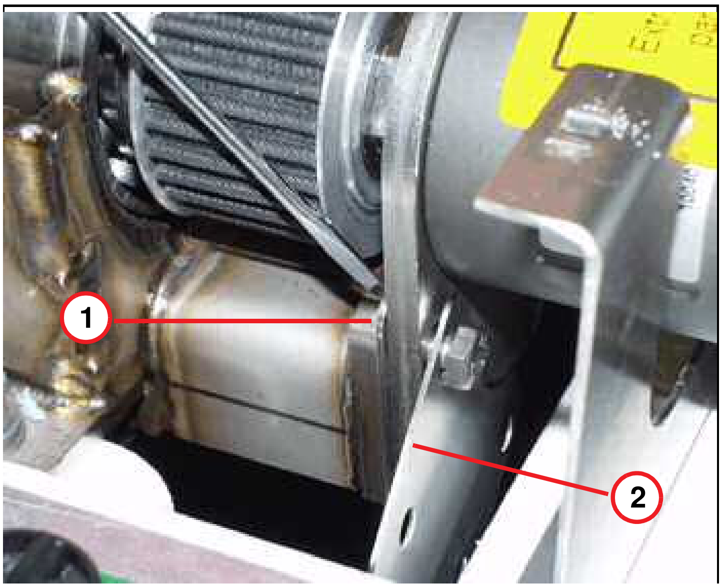
Fig. 185: M6 hex screws of MDSD bracket



(1) Shim gap appears when securing the 22 mm nut

- While torquing the 22 mm nut, a small clearance appears.

Fig. 186: Insert shim plates



(1) Shim plates to be added
(2) Spare shim plates

- Use the shim plates to fill the clearance between the motor carrier and the axle (delivered with the spare unit, or component of the original part). Tighten all four hex screws.

3.8.5.2 MDSD drive of partial drive, not valid for Trio A Tim Systems

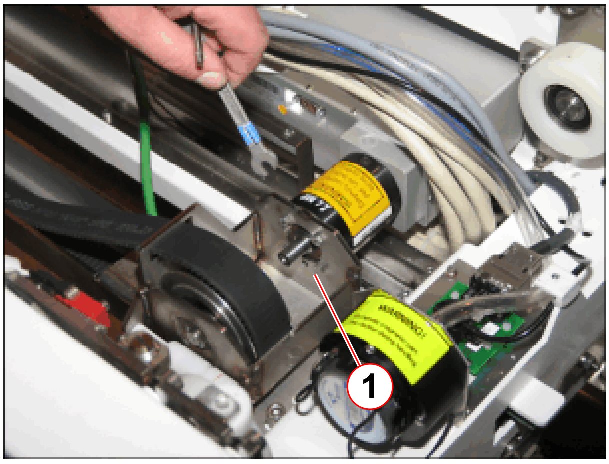
CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!**

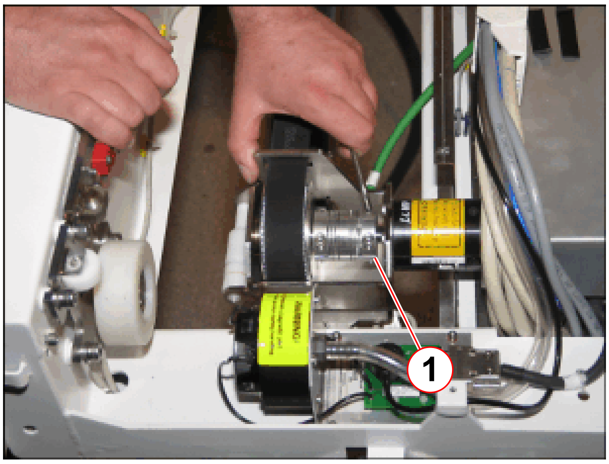
Fig. 187: MDSD clutch removed



(1) Mounting screws of MDSD

- Remove the hex screws (1) of the mounting bracket from the MDSD.

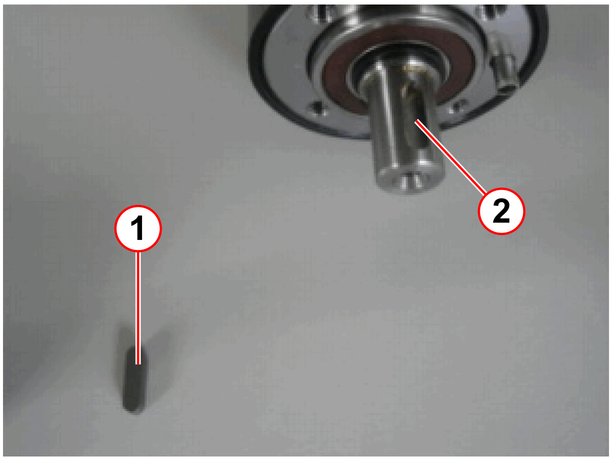
Fig. 188: *Clutch of MDSD, partial drive*



(1) Allen screws of MDSD clutch

- Remove Allen screws of MDSD clutch (1).
- Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!

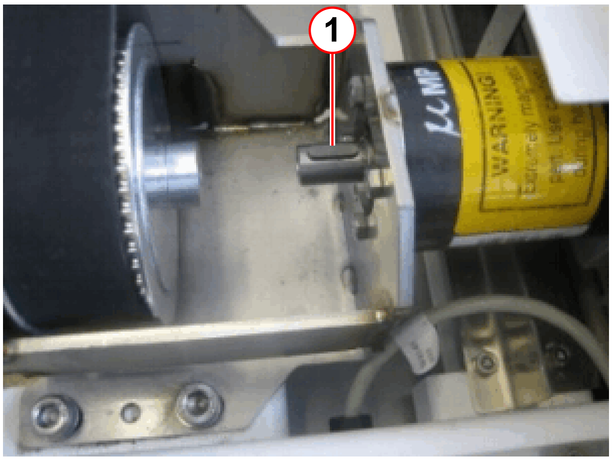
Fig. 189: *Assembling the partial drive MDSD*



(1) Key
(2) Notch of the MDSD axis

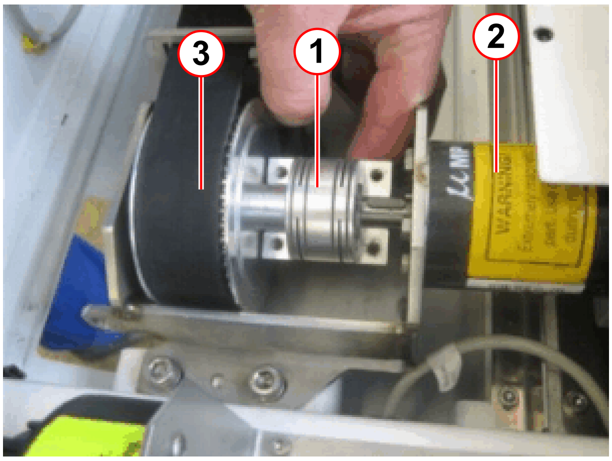
- For partial drive tables insert the key (1), delivered with the spare part, into the notch of the MDSD axis.

Fig. 190: *Key inserted in MDSD axis*



(1) Key

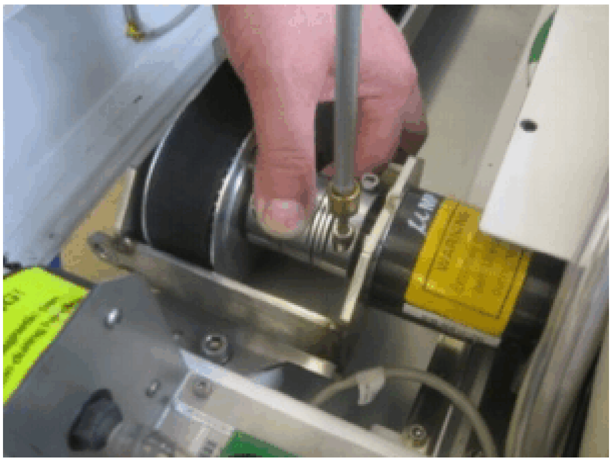
Fig. 191: Fixation of the clutch



(1) Clutch
(2) MDSD for partial drive
(3) Drive unit TK

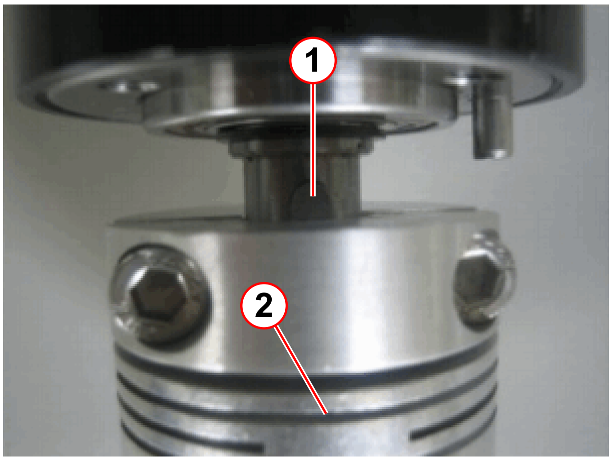
- Place the clutch as shown in pos. (1).

Fig. 192: Fixation of the clutch



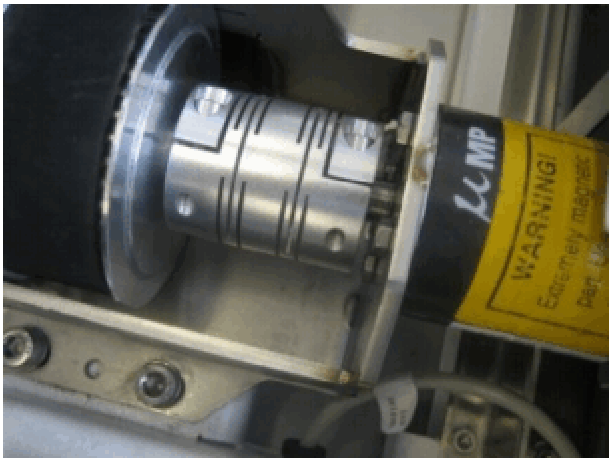
- Position the other half of the clutch so that the key fits into the notch of the clutch and tighten the screws of the clutch.

Fig. 193: Key position with clutch



(1) Key
(2) Clutch

Fig. 194: Clutch installed



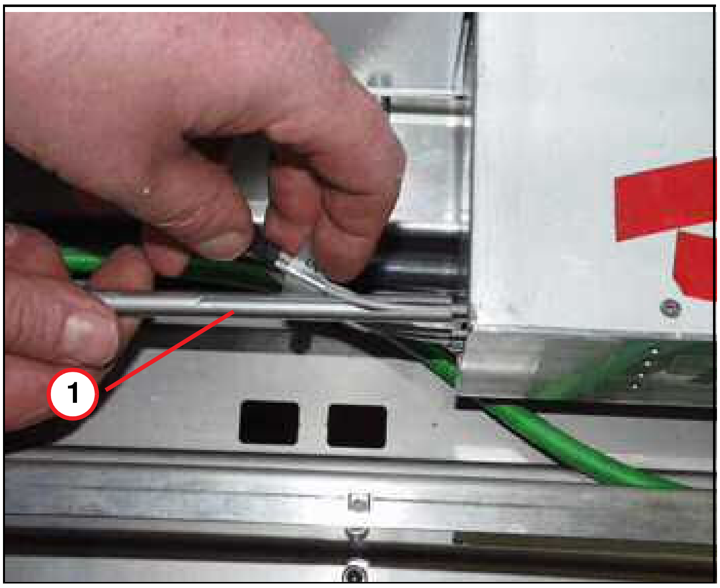
For further assembling of the partial drive, please refer to (➔ Assembly / Page 167), not valid for Trio A Tim Systems.

3.8.6 Adjustments

For adjustment of the toothed belt, please refer to (➔ Adjustments / Page 170), not valid for Trio A Tim Systems.

3.8.7 Concluding steps

Fig. 195: MDSD FOC connections



(1) FOC feed-in tool

- Reconnect the cables and FOCs on the MDSD drive and use the feed-in tool (service aid) delivered with the spare part.

- Reinstall the covers in reverse order, tighten all screws at the end of the assembling procedure.

- Switch ON F22 (RF-room).

3.8.8 Adjustments

For adjustment of the toothed belt, please refer to (→ Adjustments / Page 170), not valid for Trio A Tim Systems.

3.8.9 Final checks

- Please refer to final checks horizontal drive unit (→ Final checks / Page 157).